

Bounded Motor Intent

Request motion; observe what may actually be applied

Three LEDs make direction, enable, and reversal dead time visible.

Lesson	020 — component	Time	150–210 minutes
Prerequisites	Lessons 001–019	Board	Arduino Mega 2560
Primary evidence	TP-F, TP-R, TP-E	Status	host verified; bench open
Primary safety	E1: USB-powered LED simulation; no motor power		

1 Mission

Build a deterministic boundary between a motion request and applied driver intent:

operator request → **MotorIntent** → **MotorIntentSnapshot** → three evidence LEDs.

A request for the opposite direction never flips the direction signals while enable is active. The controller first applies stop, waits through an explicit dead-time deadline, then applies the opposite command. Stop dominates every phase. Moving intent also requires an explicit admitted **PowerDomain**; the sketch admits it only after every evidence endpoint initializes.

Primary experiment. The LEDs simulate H-bridge logic inputs. No motor, motor supply, relay, or driver power stage is connected. The exercise proves command timing; it does not prove torque, current, rotation, or motor polarity.

1.1 Learning outcomes

- distinguish requested motion from applied electrical intent;
- explain why reversal has an all-inactive interval;
- replay transitions using caller-supplied time;
- observe direction, PWM enable, acquisition, and safe state separately;
- identify what must be measured before an E2 motor continuation.

2 Parts and current

Part	Requirement
Arduino Mega 2560	USB-powered board
Three LEDs	forward, reverse, and PWM-enable evidence
Three resistors	1 kΩ, one in series with each LED
Logic analyzer	preferred for TP-F, TP-R, and TP-E timing
Multimeter	safe-state level and resistor checks; not PWM timing

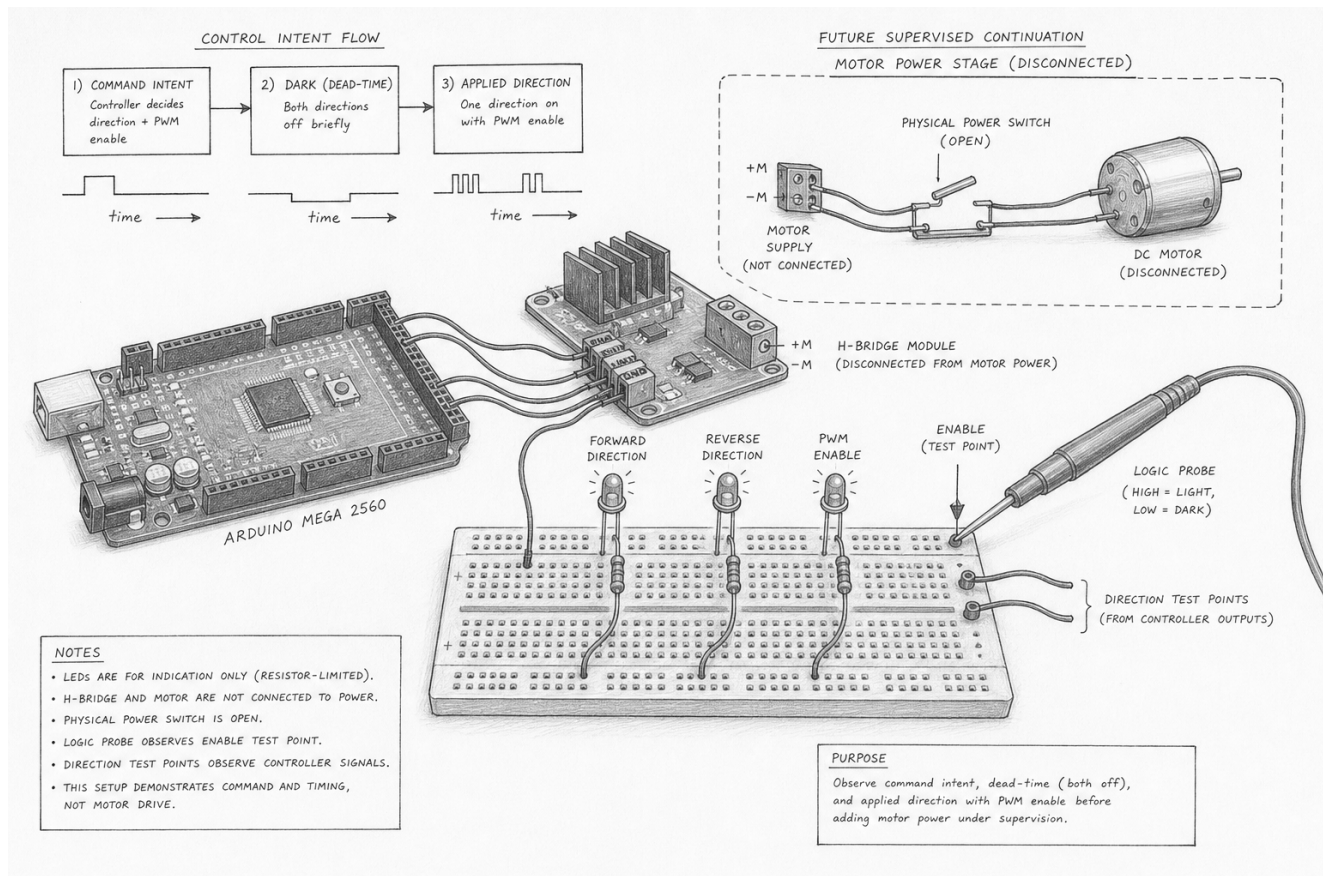
At a conservative 2 V LED drop from a measured 5 V rail,

$$I = \frac{5\text{ V} - 2\text{ V}}{1000\ \Omega} = 3\text{ mA}$$

per continuously active LED. Record the actual rail, resistor, and LED drop. Remove USB power before wiring or moving probe clips. Stop on a hot part, unstable USB connection, unexpected illumination, or a mismatch with the wiring table.

3 Authoritative wiring

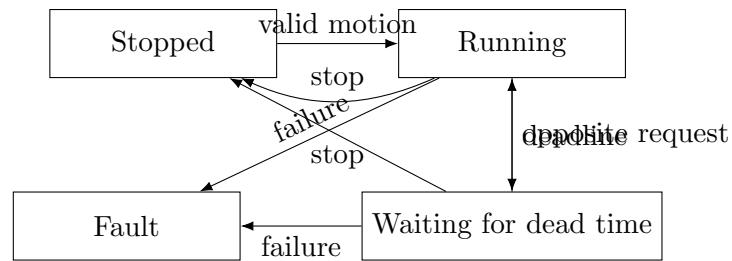
Signal	Mega pin	Connection
Forward / TP-F	D22	D22 $\rightarrow$ 1 k Ω $\rightarrow$ LED anode; cathode $\rightarrow$ GND
Reverse / TP-R	D23	D23 $\rightarrow$ 1 k Ω $\rightarrow$ LED anode; cathode $\rightarrow$ GND
Enable / TP-E	D6	D6 $\rightarrow$ 1 k Ω $\rightarrow$ LED anode; cathode $\rightarrow$ GND
Acquisition evidence	D13	built-in LED; no external wire
Probe reference	GND	common reference for TP-F, TP-R, and TP-E



Alternative text: graphite orientation plate showing a Mega and breadboard with three resistor-limited LEDs for forward, reverse, and PWM enable. A logic probe touches the enable test point. A disconnected H-bridge and motor sit inside a future supervised continuation area with an open physical motor-power switch. The table above is electrically authoritative.

The illustration is an orientation aid, not a wiring schematic. In particular, the motor and power stage remain disconnected during the supported experiment.

4 Intent and phase



Phase	TP-F	TP-R	TP-E	Meaning
Stopped	low	low	zero duty	no applied motion intent
Forward	high	low	bounded PWM	forward intent applied
Reverse	low	high	bounded PWM	reverse intent applied
Dead time	low	low	zero duty	reversal request pending
Fault	low	low	zero duty	intent cannot be trusted

No state permits both direction signals high. The first interface deliberately does not expose braking because H-bridge brake truth tables vary by device.

5 Deterministic controller

`MotorCommand` contains a direction and duty. Stopped requires duty zero; forward or reverse requires a nonzero duty no greater than the configured maximum. The snapshot preserves both requested and applied commands.

```
void loop ()
{
    const adk::TimePoint now (millis ());

    observeOperatorRequest (now);
    decideMotorIntent      (now);
    actuateMotorEvidence   ();
}
```

The canonical example introduces the runtime, evidence endpoints, power gate, then motor intent. Setup acquires, configures, and starts. The loop observes one scheduled request, advances the explicit-time decision, then writes only the *applied* snapshot.

5.1 Command rules

1. A valid stopped request applies immediately.
2. A first moving request applies its direction and bounded duty.
3. A same-direction duty change applies without dead time.
4. An opposite request applies stop immediately and records a deadline.
5. At the deadline, the pending opposite command becomes applied.
6. Stop during dead time cancels the pending motion.
7. Invalid input changes neither requested nor applied command.

Hidden calls to `millis()`, blocking `delay()`, callbacks, and heap-owned commands would make replay harder and do not belong in the controller.

6 Experiment A: acquire separately

Predict. D13 illuminates only after the controller and all three evidence outputs initialize. TP-F, TP-R, and TP-E remain inactive.

Observe. Reset the Mega. Record the D13 sequence and then measure the three test points relative to the named GND reference.

Interpret. The D13 light supports successful acquisition. Low evidence signals support the stopped electrical state. Neither observation substitutes for the other.

Signal	Prediction	Observation	Interpretation
D13	_____	_____	_____
TP-F	_____	_____	_____
TP-R	_____	_____	_____
TP-E	_____	_____	_____

7 Experiment B: direction truth table

Run the scripted stopped, forward, stopped, and reverse sequence. Predict the three signals before observing them.

Step	TP-F predicted	TP-R predicted	TP-E predicted	Observed
Stopped	low	low	zero duty	_____
Forward	high	low	PWM	_____
Stopped	low	low	zero duty	_____
Reverse	low	high	PWM	_____

An illuminated enable LED shows switching energy averaged by the eye. Use the logic analyzer at TP-E to establish duty and waveform. A direction LED proves only its logic-level evidence path.

8 Experiment C: reversal dead time

Predict. When reverse is requested during forward motion, enable becomes zero and both direction LEDs go dark immediately. Reverse becomes active only at the snapshot deadline.

Observe. Capture TP-F, TP-R, and TP-E on the same time base. Measure from the falling enable edge to the first reverse/enabling edge.

$$t_{\text{dark}} = \text{_____} \quad t_{\text{configured}} = \text{_____} \quad \text{difference} = \text{_____}$$

Interpret. A dark interval at least as long as the configured dead time supports the reversal contract. It does not measure H-bridge switching delay because no power driver is present.

9 Experiment D: stop dominance

Request stop once while running and once during dead time. In both cases, predict immediate stopped intent and no later spontaneous restart.

Starting phase	Immediate evidence	After old deadline	Pass?
Running	_____	_____	_____
Dead time	_____	_____	_____

10 Diagnosis

Observation	Likely boundary	Next unpowered check
No D13 acquisition light	claim/configuration failure	pin assignments and prior owners
Both direction LEDs active	adapter defect or wiring short	D22/D23 paths; do not connect a driver
No dark reversal interval	controller/adapter timing defect	reproduce the host trace and inspect all three channels
Direction changes but enable stays dark	duty zero, PWM claim, or D6 path	resistor, LED polarity, and D6 continuity
LED pattern correct, motor later fails	outside the LED evidence boundary	motor supply, driver rating, wiring, load, and measured current
LED dark after shutdown	safe level not high impedance	measure pin mode with the documented acceptance method

11 Deterministic test ledger

- invalid direction/duty/configuration before state change;
- stopped, first motion, same-direction duty change, and duty bounds;
- forward-to-reverse and reverse-to-forward dead time;
- just before, exactly at, and just after the deadline;
- stop and replacement requests during dead time;
- timestamp wraparound and long time jumps;
- stable snapshots and byte-identical replay;
- adapter acquisition rollback and resource reuse;
- inactive outputs during reset, fault, shutdown, and destruction.

12 CLI workflow

```
make host-test
make arduino-Lesson020MotorIntent
make upload-Lesson020MotorIntent PORT=/dev/ttyACM0
make hardware-card LESSON=020 BOARD=mega2560
make evidence LESSON=020 SIGNAL=TP-E
make lessons
make lessons-check
make site-check
```

Serial logging may record requested/applied snapshots, but it cannot replace TP-F, TP-R, TP-E, and safe-state measurements.

13 E2 continuation gate

Do not connect motor power merely because the LED experiment passes. A future supported continuation must select one H-bridge module and cite its primary datasheet. Record:

- motor stall current at the chosen current-limited supply;
- continuous and peak driver ratings, logic thresholds, flyback strategy, and thermal behavior;
- separate logic and motor-power paths plus grounding or isolation;
- a physical switch that removes motor power;
- a restrained motor or raised wheel with no pinch or entanglement path;
- measured inactive, forward, reverse, dead-time, reset, and fault states.

Stop rule. Software shutdown is not an emergency stop. Remove motor-supply power with the physical switch. USB may remain connected only if the selected module cannot back-power the Mega. Uncertainty stops the E2 continuation; it does not block the LED simulation.

14 Exercises

1. Draw the three evidence signals for a forward-to-reverse request.
2. Explain why the snapshot needs both requested and applied commands.
3. Add a same-direction duty change to the deterministic host trace.
4. Inject stop one tick before the reversal deadline and predict every later snapshot.
5. Design a separate wheel-encoder observation interface. State why commanded direction is not observed direction.

15 Bench acceptance record

Date / operator	_____
Board / revision	_____
Arduino CLI / AVR core	_____
Measured 5 V rail	_____ V
LED resistors	TP-F _____ Ω ; TP-R _____ Ω ; TP-E _____ Ω
Acquisition sequence	_____
Stopped measurements	_____
Forward measurements	_____
Reverse measurements	_____
Configured / measured dead time	_____ / _____
Stop during dead time	_____
Shutdown / reset evidence	_____
Deviations / stop conditions	_____

Status remains **hardware acceptance open** until this record is completed on a Mega 2560. The HTML reference carries current API links and corrections; this PDF carries the printable experiment and evidence ledger.